

Exercise 1

(i) Sub y_1 into y_2 : $(mx + \frac{15}{4m})^2 = 15x$
 $\Rightarrow m^2x^2 - \frac{15}{2}x + \left(\frac{15}{4m}\right)^2 = 0$

tangent $\Rightarrow b^2 - 4ac = 0$: $\left(-\frac{15}{2}\right)^2 - 4(m^2)\left(\frac{15}{4m}\right)^2 = 0$

$$\Rightarrow \frac{225}{4} - \frac{225}{4} = 0$$

m does not affect discriminant and hence $b^2 - 4ac = 0 \forall m \neq 0$

(ii) $y = mx + \frac{15}{4m}$ is tangent to $y^2 = 15x \forall m \neq 0$

So common tangents are found when $(y = mx + \frac{15}{4m})$ tangent to $(x^2 + y^2 = 16)$

Sub l_1 into l_3 : $(mx + \frac{15}{4m})^2 + x^2 = 16$

$$\Rightarrow (m^2 + 1)x^2 + \frac{15}{2}x + \left(\frac{225}{16m^2} - 16\right) = 0$$

tangent $\Rightarrow b^2 - 4ac = 0$: $\left(\frac{15}{2}\right)^2 - 4(m^2 + 1)\left(\frac{225}{16m^2} - 16\right) = 0$

$$\Rightarrow \frac{225}{4} - 4\left(\frac{225}{16} - 16m^2 + \frac{225}{16m^2} - 16\right) = 0$$

$$\Rightarrow 64m^2 + 64 - \frac{225}{4m^2} = 0$$

$$\Rightarrow 256m^2 + 256m - 225 = 0$$

$$\therefore m^2 = \frac{-256 \pm \sqrt{(256)^2 - 4(256)(-225)}}{2(256)}$$